

Introduction to data management

- Practical data management for animal health Professionals
- Session 1A: Introduction to data management
- Facilitator: John Grewar
- Date: 3 March 2025



Date	Session	Coverage
3 March 2025	1A: 08:00 – 12:00 Online contact session	Basics of data management, the importance of reliable, accurate and reproducible methods and the ethical management of data. Discussions regarding data software options fit for purposes within Animal Health.
	1B: 13:00 – 17:00 Self paced exercises with mobile access to course facilitator. 15 min sessions at 15:00 and 16:45 for group based discussion if required	Question based self learning with reading materials suggested and provided.
4 March 2025	2A: 08:00 – 12:00 Online contact session	Target EpiCollect5 with the final goal of participants understanding how to create an account, create a private data form for animal health data using data required for outbreak investigation as a backbone, capture data and have the ability to download said data from the platform.
	2B: 13:00 – 17:00 Self paced exercises with mobile access to course facilitator. 15 min sessions at 15:00 and 16:45 for group based discussion if required	Create, deploy and capture each attendees own data form on EpiCollect providing access of form to facilitator
5 March 2025	3A: 08:00 – 12:00 Online contact session	Target MS Excel. Manipulate, validate and explore animal health datasets. Establish how to aggregate and analyse data using basic statistics and graphing techniques. Finally linking excel sheets with qGIS and create a basic map of outbreak data (which will contain location data).
	3B: 13:00 – 16:00 Self paced exercises with mobile access to course facilitator. 16:00 – 17:00 Course wrap up	The facilitator will provide a dataset and some required outputs for attendees to work through.



Learning objectives

- Understand the basics of data management
- Recognize the importance of reliable, accurate, and reproducible methods
- Learn about ethical considerations in data handling
- Explore software options for animal health data management



Ice Breaker

- Each participant to put three emogi's in the Chat which they feel best describes their work in regard to Data Management

   ("Livestock vaccinations and reporting")
   ("Disease surveillance and messy data headaches")
   ("Endless paperwork and data entry struggles")



What is data management?

- Definition

The process of collecting, storing, organizing, and maintaining data

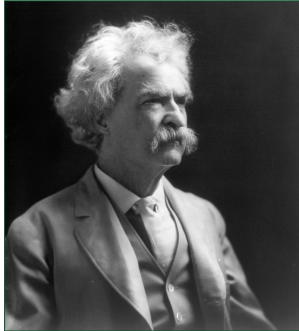
- Key components:

- Data collection
- Data storage
- Data processing
- Data analysis
- Data sharing
- Data security

Data management

the process of collecting, storing, organising, and maintaining data





*"Data is like garbage.
You'd better know what
you are going to do with it
before you collect it."*
Mark Twain



Why is data management important?

- Ensures data accuracy and reliability
- Supports decision-making in animal health
- Legal and ethical implications
- Facilitates research and outbreak investigations



Why collect epidemiologic data?

- Freedom from disease in country or region
- Prevalence and/or distribution of a disease

Received: 22 October 2019 | Revised: 3 March 2020 | Accepted: 26 March 2020
DOI: 10.1111/revd.12566

ORIGINAL ARTICLE

 WILEY

Post-outbreak African horse sickness surveillance: A scenario tree evaluation in South Africa's controlled area

John Duncan Grewar^{1,2} | Thibaud Porphyre³ | Evan S. Sergeant⁴ |
Camilla Theresa Weyer^{2,5} | Peter Neil Thompson¹

Article — Artikel

Descriptive study of an outbreak of equine sarcoid in a population of Cape mountain zebra (*Equus zebra zebra*) in the Gariep Nature Reserve

P. J. Nel⁶, H. Bertschinger⁶, J. Williams⁶ and P. N. Thompson⁶



Why collect epidemiologic data?

- Risk assessment
 - Release
 - Exposure
 - Consequence
- Risk factors

OPEN ACCESS PEER-REVIEWED
RESEARCH ARTICLE

Quantitative Risk Assessment for African Horse Sickness in Live Horses Exported from South Africa

Evan S. Sergeant¹, John D. Grewar², Camilla T. Weyer³, Alan J. Guthrie⁴
Published: March 17, 2016 • <https://doi.org/10.1371/journal.pone.0151757>

PLOS COMPUTATIONAL BIOLOGY

RESEARCH ARTICLE

Modelling African horse sickness emergence and transmission in the South African control area using a deterministic metapopulation approach

Joanna N. de Klerk^{1*}, Erin E. Gorsich¹, John D. Grewar², Benjamin D. Atkins³, Warren S. D. Tennant⁴, Karen Labuschagne⁵, Michael J. Tildesley⁶



Journal of Food Protection
Volume 47, Issue 5, 1 May 2004, Pages 980-992



Public Health Consequences of Macrolide Use in Food Animals: A Deterministic Risk Assessment[†]

H. Scott Hurd^{1,2,3}, Stephanie Doores², Dermot Hayes¹, Alan Mathew⁴, John Maurer⁵, Peter Silley⁶, Randall S. Singer⁷, Ronald N. Jones⁸



Why collect epidemiologic data?

- Risk factors

The landscape epidemiology of foot-and-mouth disease in South Africa: A spatially explicit multi-agent simulation

Elise Dion^{a,*}, Louis VanSchalkwyk^b, Eric F. Lambin^{a,c}

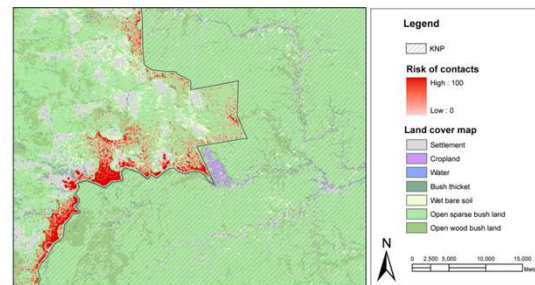


Fig. 7. Map of high contact areas showing the presence of buffaloes outside the park.



Why collect epidemiologic data?

- Identify treatment or control options for a disease
- Set priorities for the use of resources
- Assist in planning, implementing and monitoring disease control/eradication programs

OPEN ACCESS PEER REVIEWED
RESEARCH ARTICLE

Modelling the factors affecting the probability for local rabies elimination by strategic control

Johann L. Kotzé, John Duncan Greener, Aaron Anderson

Version 2 Published: March 4, 2021 • <https://doi.org/10.1371/journal.pntd.0008236>

Model parameter ^a	Parameter space	Comments
population size	(200, 400, 1,800, 1,400, 16,200) Set = 5	The initial population size at the start of the iteration. The parameter progressively increases from 200 to 16200 by a factor of 5. The default value for calculations was chosen as 5,400 (closest to the mean of 1,400).
infectivity	(0.5 ≤ x ≤ 1.5) Set = 5	A factor that is multiplied with the rate of effective bites (0.02252 bites per rabid animal per day). An effective bite is infective and implies infectivity. The default value is 1.
fecundity	(0.5 ≤ x ≤ 1.5) Set = 5	A factor that is multiplied with the default mean litter size (4.4) and the default mean litters per year for each female female (0.31). The default value is 1.
in-migration	(0 ≤ x ≤ 0.1) Set = 5	The in-migration allowed per annum as a fraction of the current population size. The chosen values are lower than the published value of 0.28. This is because high in-migration rates would substantially reduce the ability to observe the effects of the other parameters. In-migrants are assumed immune naïve and not infective. The default value was chosen as 0.
exposure survival	(0 ≤ x ≤ 0.001) Set = 5	The probability for an effectively contacted dog to immune-convert without overt disease. The published value is zero as this is not currently considered an important factor in rabies epidemiology. Low values were chosen to investigate the possible importance of such occurrences (if true). The default value was chosen as 0.
initial incidence	(0.005, 0.01, 0.05) Set = 3	The epidemic is started by introducing for 2 consecutive months infected dogs equal to the annualised incidence. The maximum annual incidence (0.01) is twice reduced by a factor of 2 to produce the parameter space. The default value was chosen as 0.01.
initial immunity	(0 ≤ x ≤ 1) Set = 11	The herd immunity at the start of each simulation (as a result of vaccination). Vaccinations are assumed to be 100% effective. All possible levels of herd immunity were investigated. The default value was chosen as 0.3 (a value typical of low intervention populations).

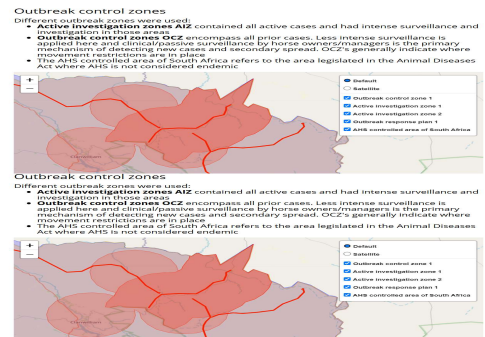
^a These model parameters were also used as the independent variables in the regression model.

<https://doi.org/10.1371/journal.pntd.0008236.t001>



Why collect epidemiologic data?

- Assist in planning, implementing and monitoring disease control/eradication programs



- Meet international reporting responsibilities
- Demonstrate disease status to trade partners

[illegible]


World Organisation for Animal Health | WAHIS

[Analysis](#) > [Reports](#) > [Alerts](#) > [Data](#) > [Tools](#) > [Help](#)

10/20/2016

1. Select the report page

Six-monthly report on the notification of listed diseases, infections and infestations

Final semester 2016 - South Africa - [View report](#)

[Download PDF](#)

Search in subcategory

Report	Subcategory of
6/1/2016 - 1/1/2017 6/1/2015 - 1/1/2016 6/1/2014 - 1/1/2015 6/1/2013 - 1/1/2014 6/1/2012 - 1/1/2013	South Africa

World Organisation for Animal Health WAHIS

Menu

EVENTS

EVENTS 2019

Workshop: African swine fever virus (ASFV)

Event name	Event dates	Event location	Event dates
African swine fever virus (ASFV) workshop	2019-01-15 to 2019-01-17	Geneva, Switzerland	2019-01-15 to 2019-01-17



Why collect epidemiologic data?

- Evaluate diagnostic tests



Preventive Veterinary Medicine
Volume 233, December 2024, 106358



Assessment of sensitivity and specificity of bacterial culture and the VetMAX™ MastiType Multi Kit in detecting *Streptococcus uberis* and *Escherichia coli* in milk samples from dairy cows with clinical mastitis in subtropical Australia

Tasneem Imam ^{a,1}, Sara Horsman ^a, Ben Wood ^a, John D. Grewar ^b,
Charlotte Langhorne ^b, Rochelle Price ^a, Caitlin Wood ^a, Joerg Henning ^{a,2}, Justine S. Gibson ^{a,2}





Common data management challenges

- Data inconsistency and missing values
- Lack of standardised formats
- Data security risks
- Difficulty in integrating data from different sources



Ethical management of data

- Ownership and data privacy
- Informed consent and confidentiality
- Compliance with regulatory frameworks (e.g., WOH, national regulations)



Data software options for animal health

- Spreadsheet-based tools
 - Microsoft Excel, Google Sheets
- Database tools
 - PostgreSQL, Microsoft Access
- Mobile data collection
 - EpiCollect5, OpenDataKit (ODK)
- GIS tools
 - QGIS, ArcGIS
- Statistical analysis
 - R, Python, Proprietary programs



Selecting the right software

- Considerations when choosing a tool
 - Accessibility
 - Scalability
 - Cost
 - Ease of use
- Case examples of software fit for different purposes



Key components

- Data collection
- Data storage
- Data processing
- Data analysis
- Data sharing
- Data security



Data collection

Data collection

The process of gathering data from various sources

- Key considerations
 - Defining objectives and data requirements
 - Ensuring data accuracy and completeness
 - Choosing the right tools (e.g., surveys, sensors, field observations)



Data collection

- Examples
 - Outbreak investigations
 - Lab submissions
 - Disease reporting

[illegible][illegible]

Directorate: Animal Health SR1				agriculture, forestry & fisheries
Notification of an Animal Disease Outbreak				
Disease Name:				
Report Number:		Lab. Number:		
Date of this report:				
Date of first detection:				
Estimated date of first infection:				
Location:				
Province:				
Local Municipality:				
SV Area:				
Geographical Coordinates:		East (degrees, Minutes, Seconds)	South (degrees, Minutes, Seconds)	
Farm Name:				
Farm Number:				
Owner Name:				
Owner contact number:				
Outbreak Details:				
Species:		Diseases:		
Outbreaks:		Deaths:		Destroyed:
Contacted Population at risk:		Estimated radius affected by outbreak:		
Nature of Diagnosis:				
Name of Laboratory:				
History / Epidemiological Comments:				
Control Measures taken to date:				
Information supplied by: (name & address)				
Contact details of official:				
Fax				
E-mail				
Cell phone				
Signature: Date Information Provided Official				



Data collection

- Examples
 - Active surveillance
 - Animal movement

✕ QUIT
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NEXT >

HOLDING

Sample event date

Format: YYYY/MM/dd

* This field is required

📅 24 Feb 2025
✕

The date selected is: 2025/02/24

1 PERMIT APPLICATION FORM FOR MOVEMENT INTO THE AHS CONTROLLED AREA: 2012 v2

• This application must be completed with the AHS Permit and an AHS Certificate
 • This application must be accompanied by a copy of the 10 page and AHS restriction pages of a validated, unexpired permit for each horse travelling
 • An international permit, for use signatures, will be used for the international section below (if and when the permit is issued - this is NOT used for a permit horse travel)
 • The appropriate health certificate must be in each passport not more than 10 days prior to movement and must confirm the requirements as stated in the permit
 • **Declaration:** Please complete all of and send to: movements@ahs.gov.au

*** Section 1: Stop Over Declaration (only if required)**

• Please advise details on the each controlled area you stop up to a maximum of 7 working days to arrival
 • For each stop, please provide the following information:

1	Travelling horses	Travelling horse and ID	
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- < PREV [] NEXT >
- HOLDING**
- Sample event date**
- Format: YYYY/MM/dd
- * This field is required
- 24 Feb 2025
- The date selected is: 2025/02/24

[illegible]

Data collection

- Examples
 - Regulatory work

Western Cape Government **AHS Field Regulation** Event
Form Version 2016-10-20

holding name		Official Name	
Address		Date of inspection	
Location: +EAST -SOUTH		Notes for officials: 1. The checks are for the day in question, not for future compliance. 2. TCA = checked and compliant 3. OHSIS = checked and not compliant - put details in comments section 4. NA = Not applicable to be checked 5. Try not to leave fields blank since this is difficult to interpret	
Owner:		Transport Details (if applicable)	
Telephone: Email:		Vehicle Registration	
Cell Phone: Private: Vet:		Transporter company	
		Destination	

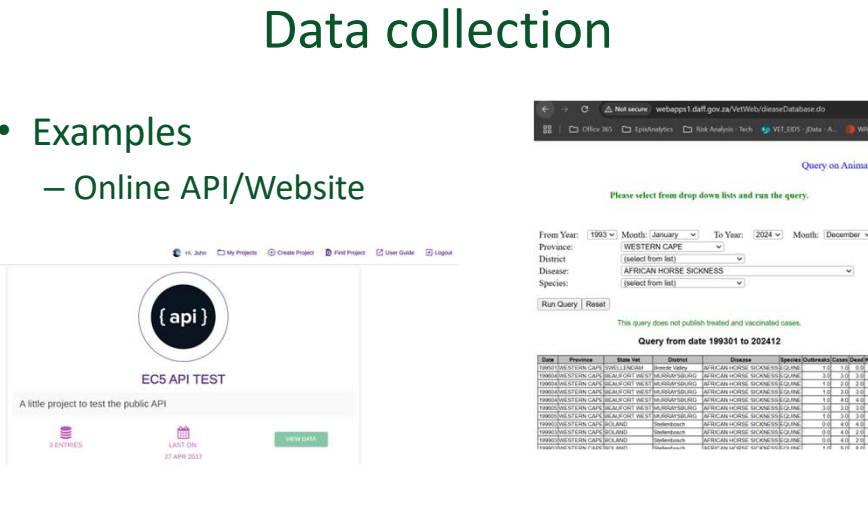
Horse name	Microchip #	Passport #	Breed	Sex male female unknown	Date of Birth MM/YY	AHS 1 Date	status	private vet	AHS 2 Date	status	private vet	Permit Valid	Health Cert	OWNER

Please indicate any issues with specific horses



Data collection

- Examples
 - Online API/Website



The image displays two examples of online data collection interfaces. On the left is the 'EC5 API TEST' website, which features a navigation bar with links like 'Home', 'About', 'My Projects', 'Create Project', 'Find Project', 'User Guide', and 'Logout'. The main content area has a large circular logo with '{ api }' inside, the text 'EC5 API TEST', and a description: 'A little project to test the public API'. Below this, it shows '3 ENTRIES' and 'LAST ONE 27 APR 2017' with a 'view entry' button. On the right is a screenshot of a web browser showing a 'Query on Animal Diseases in the RSA' interface. It includes a search bar, a 'Please select from drop down lists and run the query.' instruction, and a form with fields for 'From Year' (1993), 'Month' (January), 'To Year' (2024), 'Month' (December), 'Province' (WESTERN CAPE), 'District' (selected from list), 'Disease' (AFRICAN HORSE SICKNESS), and 'Species' (selected from list). A 'Run Query' button is present. Below the form, a note states 'This query does not publish treated and vaccinated cases.' and a title 'Query from date 199301 to 202412' is shown. A table of results follows, with columns: Date, Province, Date Vet, District, Disease, Species/Outbreaks, Cases/Dead/Killed. The table contains 10 rows of data.

Date	Province	Date Vet	District	Disease	Species/Outbreaks	Cases/Dead/Killed
1993-01-11	WC	1993-01-11	Worcester	AFRICAN HORSE SICKNESS	AFRICAN HORSE SICKNESS	1/0/0/0
1993-01-11	WC	1993-01-11	Worcester	AFRICAN HORSE SICKNESS	AFRICAN HORSE SICKNESS	1/0/0/0
1993-01-11	WC	1993-01-11	Worcester	AFRICAN HORSE SICKNESS	AFRICAN HORSE SICKNESS	1/0/0/0
1993-01-11	WC	1993-01-11	Worcester	AFRICAN HORSE SICKNESS	AFRICAN HORSE SICKNESS	1/0/0/0
1993-01-11	WC	1993-01-11	Worcester	AFRICAN HORSE SICKNESS	AFRICAN HORSE SICKNESS	1/0/0/0
1993-01-11	WC	1993-01-11	Worcester	AFRICAN HORSE SICKNESS	AFRICAN HORSE SICKNESS	1/0/0/0
1993-01-11	WC	1993-01-11	Worcester	AFRICAN HORSE SICKNESS	AFRICAN HORSE SICKNESS	1/0/0/0
1993-01-11	WC	1993-01-11	Worcester	AFRICAN HORSE SICKNESS	AFRICAN HORSE SICKNESS	1/0/0/0
1993-01-11	WC	1993-01-11	Worcester	AFRICAN HORSE SICKNESS	AFRICAN HORSE SICKNESS	1/0/0/0
1993-01-11	WC	1993-01-11	Worcester	AFRICAN HORSE SICKNESS	AFRICAN HORSE SICKNESS	1/0/0/0



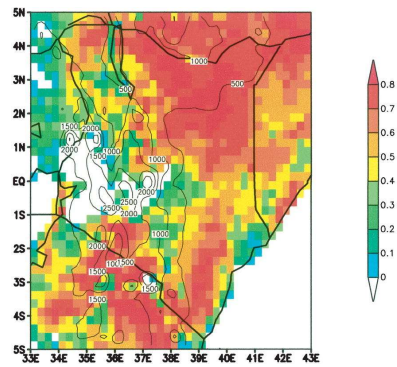
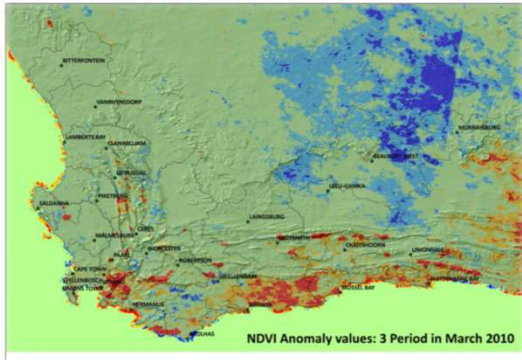
The JDATA logo is located in the bottom right corner of the image. It consists of a stylized 'J' in green and the word 'DATA' in black, all enclosed within a black rectangular border.

- 

[illegible]

Data collection

- Examples
 - Remote sensing



Data storage

- Process of storing digital information for future access
- Essential for animal health records, outbreak reports, and laboratory data
- Prevents data loss and ensures compliance with regulations

Data storage

Ensuring secure, organised, and accessible storage of collected data



Key considerations

- **Data security:** Encryption, access control, backup strategies
- **Data format & structure:** Standardisation (CSV, JSON, SQL) & structured vs. unstructured data
- **Scalability:** Ability to handle increasing data volumes
- **Regulatory compliance:** GDPR, POPI as examples



Types of storage systems

- **Where?**

- **Local** storage: Hard drives, USB, on-premise servers
- **Cloud** storage: Google Drive, AWS S3/RDS, Microsoft OneDrive
- **Hybrid** Storage: Combination of local & cloud storage

- **How?**

- Relational databases (SQL): PostgreSQL, MySQL, Microsoft SQL Server, MS Access
- NoSQL Databases: MongoDB, Firebase, DynamoDB
- Microsoft Excel



Advantages and disadvantages

- **Local storage (On-premises)**

- **Pros:** Full control, no internet dependency.
- **Cons:** Risk of data loss, limited scalability
- **Example:** Veterinary clinics storing Excel-based vaccination records

- **Cloud storage**

- **Pros:** Scalable, accessible from anywhere, automatic backups
- **Cons:** Requires internet, potential privacy concerns
- **Example:** Animal disease surveillance data stored in WAHIS database for real-time access



Data storage best practice

- **Use multiple backups:** Local + cloud + external copies
- **Encrypt sensitive data:** Protect confidential information (or simply don't capture it if possible)
- **Automate storage processes:** Use APIs to fetch & store data
- **Role-based access control:** Restrict sensitive data access



Demo

PostgreSQL
API
Collaboration



Summary Storage

Storage type	Best for	Pros	Cons
Local storage	Small datasets	Easy, no internet needed	Data loss risk
Cloud storage	Collaborative work	Scalable, accessible	Privacy concerns
SQL databases	Large, structured data	Fast queries, relationships	Requires expertise
NoSQL	Unstructured, real-time data	Flexible, big data	Not for complex relationships
Hybrid storage	Secure & accessible data	Balances security & usability	Needs synchronization – EpiCollect5



Planning for 1500 and 1645

- BREAK until 10:45
- Accessing WhatsApp
- 15:00
 - John Grewar is inviting you to a scheduled Zoom meeting. Join Zoom Meeting <https://us02web.zoom.us/j/81269598340?pwd=yafsYV5RrM41sa2g91zaXzLJ3hcadM.1>
 - Meeting ID: 812 6959 8340 Passcode: 545195
- 16:45
 - <https://us02web.zoom.us/j/85440548523?pwd=V634RlWjWvmC27zYF4rVrjkJaRPHXf.1>
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Data processing

- Converting collected data into a structured format
- Includes
 - cleaning
 - validation
 - transformation
 - analysis
- Ensures accuracy, consistency, and usability of data

Data processing

Transforming raw data into meaningful, clean data to turn into information



Data processing

- Data cleaning
 - Removing errors, duplicates, and inconsistencies
- Data transformation
 - Standardising formats, merging sources
 - Example
 - Taking multiple case definition outputs to classify case or non-case status
- Data validation
 - Ensuring logical consistency and completeness
- Data analysis
 - Aggregation, filtering, and identifying patterns



Data processing

- Spreadsheet-based
 - Microsoft Excel, Google Sheets (basic processing)
- Programming languages:
 - R (*dplyr*, *tidyr*)
 - Python (*pandas*, *NumPy*)
- Database queries
 - SQL (data extraction and transformation)



Data processing

- Remove duplicates using 'Remove Duplicates' feature.
- Use filters to check for missing values.
- Apply text-to-columns for splitting data.
- Use pivot tables for quick summarization



Data processing



Demo

Remove duplicates
Use filters to check for missing values
Splitting data
Summaries to do basic checks



Data analysis

- The process of examining, transforming, and modeling data to extract insights
- Helps in identifying trends, patterns, and relationships in data
- Supports evidence-based decision-making in animal health and epidemiology

Data analysis

Turning processed data into meaningful information



Types of data analysis

- **Descriptive analysis**
 - Summarising and visualising data (mean, counts, trends)
- **Exploratory analysis**
 - Detecting patterns and outliers in datasets
- **Inferential analysis**
 - Using statistical methods to draw conclusions
- **Predictive analysis**
 - Using models to forecast future trends



Basic techniques for data analysis

- **Summarisation**
 - Calculating averages, counts, and proportions
- **Data visualisation**
 - Creating charts, histograms, and scatter plots
- **Comparisons**
 - Examining relationships between variables (e.g., correlation)
- **Filtering & grouping**
 - Aggregating data to extract insights from subgroups



Common tools for data analysis

- **Spreadsheet-based**
 - Microsoft Excel (PivotTables, charts) – see later in course
- **Programming Languages**
 - R (*ggplot2*, *dplyr*)
 - Python (*pandas*, *matplotlib*, *seaborn*)
- **Statistical software** (if not above)
 - SPSS, Stata (advanced statistical modeling)
- **GIS tools** (if not above)
 - QGIS, ArcGIS (spatial data analysis)



Why data analysis matters

- Helps in disease **surveillance** (e.g., detecting outbreaks early)
- Supports **risk assessments** for animal health interventions
- Provides **evidence-based insights for policy** and decision-making
- Essential for **research** and **reporting**



Data analysis



Demo

Summary Statistics

Counts

Plots

Basic stats – Chi Square/Fishers exact



Data sharing

- The process of making data available to relevant stakeholders
- Enables collaboration, transparency, and informed decision-making
- Can be done via API's, websites, reports, dashboards, or cloud storage

Data sharing

ensuring data is accessible, secure, and usable for decision-making and collaboration



Methods of data sharing

- **API's** (application programming interface)
 - Automated data exchange between systems
- **Online dashboards**
 - Real-time visualizations (Tableau, Power BI, Shiny)
- **Web-based reports**
 - Static or interactive reports shared online
- **Cloud storage & file sharing**
 - Google drive, AWS S3, OneDrive, MS Sharepoint



Key considerations in data sharing

- **Data security**
 - Ensure encryption and access control
- **Regulatory compliance**
- **Data integrity**
 - Ensure accurate and up-to-date data
- **Format & compatibility**
 - Use standard formats (CSV, **JSON**, XML)



Data sharing



Demo

Dashboard
Static/Dynamic report
API



Data security

- The practice of protecting digital information from unauthorized access, corruption, or theft
- Ensures data remains confidential, available, and accurate
- Important for compliance with regulations

Data security

Ensuring the protection, integrity, and confidentiality of data in storage and transmission



Key threats

- **Unauthorized access**
 - Data breaches, hacking, phishing attacks
- **Data corruption**
 - Loss due to system failures, malware, or human error
- **Insider threats**
 - Employees mishandling or leaking data
- **Lack of encryption**
 - Data interception during transmission



Best practices for data security

- Encryption
 - Protect data in transit and at rest (AES-256, SSL/TLS)
- Access control
 - Use role-based access and two-factor authentication
- Regular backups
 - Store copies in secure locations
- Audit logs
 - Track who accessed or modified data



Data security



Demo

API key – epiCollect5 python

2FA

Encryption of dashboard/websites



Discussion & Q&A

- Open discussion on participant experiences
- Questions & insights
- jdgrewar@gmail.com



Questions & insights

1. In the day-to-day work you do what data do you collect with regards to animal health?
2. From the reasons we discussed with regards to epidemiologic data, which fit into the data you collect?
3. How clearly is the reason for data collection described
4. Are there data you feel is missing from the data collection programs you have
5. And the converse, is there data that you collect that you are not aware is analysed? What would be the downside of removing the need to capture these data?
6. Are there any other data management challenges that are pertinent to your systems?
7. What software tools are commonly used for your data management systems?



Questions & insights

8. Data collection
 1. What mobile tools do you use in the field to collect data?
 2. What are some of the advantages/disadvantages of paper-based field forms?
 3. In what format do you get data when you request it from data managers?
9. Data storage
 1. What databases/excel sheets/other tools are used to store data you collect?
 2. What is your preferred storage means
 3. Do you store, and if so how, any GIS data?
 4. Between local, hybrid and cloud-based approaches which is most used in your organisation?
 5. Do you access animal health/laboratory data from outside of your organisation? How easy/difficult is it to access this?
10. Data processing
 1. How often, and in what circumstances, do you find yourself needing to process data?
 2. What are the most common mistakes you find you/your colleagues/your employees make when it comes to data they have captured?
 3. For processing what are your go-to programs that you use? Have you tried any others we've mentioned in the course?
 4. Have you ever written a SQL query?



Questions & insights

11. Data analysis
 1. Between descriptive, exploratory, inferential and predictive which typically describes best your analysis process?
 2. Have you ever made a pivot table/so, and can you remember under what circumstance?
 3. What is your favourite graph/graph type?
 4. Have you used GIS tools before?
12. Data sharing
 1. Have you ever needed to share data in real time with colleagues and if so, how have you gone about it?
 2. Have you ever used an API?
 3. What dashboard technology do your systems use if any?
 4. What cloud service provider do your systems generally prefer?
 5. Have you ever worked with JSON or XML data?
13. Data security
 1. Have you ever been involved with dealing with a data breach?
 2. Have you ever deleted data in error and been unable to retrieve it?
 3. Of the threats we discussed which is most likely to be an issue in your environment?



Group discussions and planning

- 1300 - 1600: Optional discussion slots – WA Facilitator
- 1600 - 1700: Wrap up session
- Planning
 - Make a EpiCollect5 user at <https://five.epicollect.net/>
 - If you do have a Google account use that – its easier
 - Download the EpiCollect5 from your App Store/Play Store
 - Download qGIS at <https://qgis.org/download>

